Tutorial 6

Getting started with SPSS or PSPP

An easy video tutorial on:

SPSS https://www.youtube.com/watch?v=ZpwZS3XnEZA

PSPP https://www.youtube.com/watch?v=NqS6DTXeXC4

Activity 1

RODENTS IN SPACE: A SIMPLE MEMORY EXPERIMENT

Twenty one first-year undergraduates participated in a simple memory experiment designed to investigate the effect of a mnemonic strategy upon memory for paired words. The participants were randomly divided into two groups. All participants were given 2 minutes to memorise a list of 20 words presented in pairs. After learning the words for 2 minutes the participants were then required to recall any of the words that they could remember. The groups were split into a mnemonic group (participants were advised to try to form a mental image to link to the words) and a non-mnemonic group. The data are given below:

Memory scores (out of 20) for the mnemonic group:

20, 18, 14, 18, 17, 11, 20, 18, 20, 19, 20

Memory scores (out of 20) for the non-mnemonic group:

10, 20, 12, 9, 14, 15, 16, 14, 19, 12

Using these data to attempt the following:

1. Form a research question, which examines the difference between the two groups.
   RQ: Does the use of mnemonics affect memory? Or
   RQ: “To what extent does the utilization of mnemonics enhance memory retention in a controlled experimental setting?"
2. Form a hypothesis, which examines the difference between the two groups.
   H_0: There is no significant difference in the memory recall means between a group that used mnemonics and a group that did not use mnemonics to recall a list of words.

H_a: There is significant difference in the memory recall means between a group that used mnemonics and a group that did not use mnemonics to recall a list of words.

3. What is the dependent variable?

The memory scores

4. What is the independent variable?

The scores of the mnemonic group and the non-mnemonic group

The participants were randomly divided into two groups.
So it is a **between study** and could be studies with independent sample t test

5. In SPSS or PSPP, set up a data file to record the above data. More specifically:
   - In **Variable View** (where we set up the independent variable and dependent variable):

Look at Tutorial5.sav

     - In Name, give an appropriate Name to each variable.
     - In Type, insert Numeric for both variables
     - In Width, give an appropriate width to both variables.
     - In Decimals, type 0 for both variables.
     - In Label, give appropriate label to each variable.
     - In Values, apply a label and value to the variable which is responsible for splitting the sample.
     - In Measure, select Scale (which is either interval or ratio). Avoid using a nominal or ordinal variable as a dependent variable.
     - Regarding the rest variable properties, leave them as they are.
   - **In Data View** (where we set up the observations for each variable), insert the above observations (data) in the column of the appropriate variable.

Look at Tutorial5.spv

     - Save the .sav file, as 'MemoryStudy'.
     - Select Analyze > Descriptive Statistics > Frequencies, select both variables, select Statistics and select all the following options: Mean, Median, Mode, Std. deviation, Variance, Range, Minimum, Maximum, Skewness and Kurtosis. In Charts select Histograms and show normal curve and histogram. Look at the generated table, consult your lecture notes and discuss the output.

Statistics

		Mnemonic	Non_mnemonic
N	Valid	11	10
	Missing	10	11
Mean		17.73	14.10
Median		18.00	14.00
Mode		20	12[a]

Std. Deviation		2.867	3.573
Variance		8.218	12.767
Skewness		-1.595	.337
Std. Error of Skewness		.661	.687
Kurtosis		2.214	-.590
Std. Error of Kurtosis		1.279	1.334
Range		9	11
Minimum		11	9
Maximum		20	20
Percentiles	25	17.00	11.50
	50	18.00	14.00
	75	20.00	16.75

a. Multiple modes exist. The smallest value is shown

Frequency Table

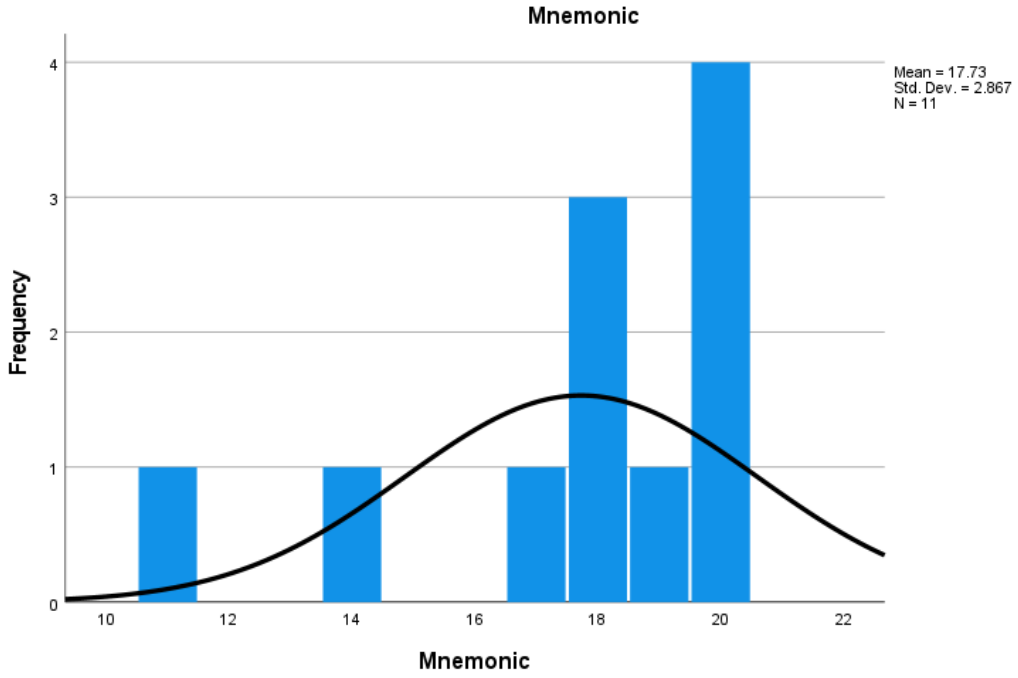
Mnemonic

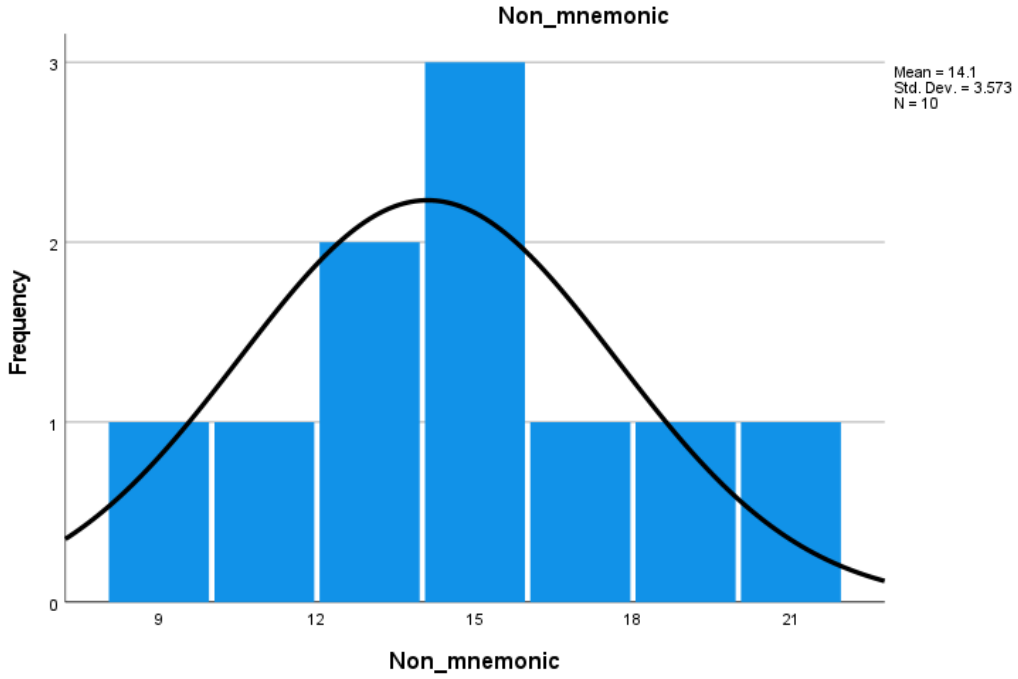
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	11	1	4.8	9.1	9.1
	14	1	4.8	9.1	18.2
	17	1	4.8	9.1	27.3
	18	3	14.3	27.3	54.5
	19	1	4.8	9.1	63.6
	20	4	19.0	36.4	100.0
	Total	11	52.4	100.0	
Missing	System	10	47.6		
Total		21	100.0		

Non_mnemonic

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	9	1	4.8	10.0	10.0
	10	1	4.8	10.0	20.0
	12	2	9.5	20.0	40.0
	14	2	9.5	20.0	60.0
	15	1	4.8	10.0	70.0
	16	1	4.8	10.0	80.0
	19	1	4.8	10.0	90.0
	20	1	4.8	10.0	100.0
	Total	10	47.6	100.0	
Missing	System	11	52.4		
Total		21	100.0		

Histogram





- Answer the following questions:
  - Do we split the sample in this type of study according to treatment?

Yes

- Is this type of study a 'between-subjects design' or a 'within-subjects design'? To answer the last question, perform research using 'Google' regarding the terms: 'between-subjects design' and 'within-subjects design'.

Between-subjects

Interpretation

1. **Central Tendency (Mean, Median, and Mode)**:
    - The **mean** for the Mnemonic group (17.73) is higher than for the Non-mnemonic group (14.10), indicating that, on average, the Mnemonic group scores higher.
    - The **median** values are 18.00 for Mnemonic and 14.00 for Non-mnemonic, consistent with the mean, suggesting the Mnemonic group tends to perform better.
    - The **mode** is 20 for the Mnemonic group and 12 for the Non-mnemonic group, with the Non-mnemonic group having multiple modes.
2. **Dispersion (Standard Deviation, Variance, and Range)**:
    - The **standard deviation** is higher in the Non-mnemonic group (3.573) compared to the Mnemonic group (2.867), indicating more variability in scores for the Non-mnemonic group.
    - **Variance** follows a similar pattern, with the Non-mnemonic group showing more spread in the data (variance = 12.767) than the Mnemonic group (variance = 8.218).
    - The **range** of scores is slightly broader for the Non-mnemonic group (range = 11) than the Mnemonic group (range = 9).
3. **Skewness and Kurtosis**:
    - **Skewness**: The Mnemonic group has a negative skewness (-1.595), suggesting a left-skewed distribution (a concentration of scores on the higher end). The Non-mnemonic group has a slight positive skewness (0.337), indicating a fairly symmetric distribution with a slight concentration on the lower end.
    - **Kurtosis**: The Mnemonic group has a positive kurtosis (2.214), indicating a leptokurtic distribution (more peaked), while the Non-mnemonic group has a negative kurtosis (-0.590), indicating a platykurtic distribution (flatter).
4. **Percentiles**:
    - The **percentiles** indicate that a higher percentage of Mnemonic group participants scored near the upper range (e.g., the 75th percentile is 20.00), whereas the Non-mnemonic group's scores are more evenly distributed.

Overall Insights

- The **Mnemonic group** generally scores higher and has less variability, with scores clustered around the upper end of the scale.
- The **Non-mnemonic group** shows greater variability, with a broader range of scores and a more balanced distribution, though tending slightly toward lower scores.

These descriptive statistics suggest that the Mnemonic technique may have a positive effect on performance, as indicated by higher average scores and a more concentrated score distribution.

Interpretation of the graphs

Mnemonic Group Histogram

- **Shape**: The histogram for the Mnemonic group would likely be **left-skewed** (negatively skewed), given the skewness of -1.595. This means the bulk of the scores would be concentrated toward the higher end, with a tail extending to the left.
- **Distribution**: Since the mode is 20 and the mean is 17.73, we'd expect more scores in the 18–20 range, creating a peak toward the right side of the histogram.
- **Spread**: With a standard deviation of 2.867, the spread of the bars would show moderate variability around the mean, with fewer scores extending down to the lower end (closer to the minimum score of 11).
- **Kurtosis**: The positive kurtosis value (2.214) indicates a **leptokurtic distribution**, meaning the histogram would have a relatively sharper peak around the mode (20) compared to a normal distribution.
- **Interpretation**: The histogram would show a concentration of higher scores, indicating that most participants in the Mnemonic group performed well.

Non-mnemonic Group Histogram

- **Shape**: The histogram for the Non-mnemonic group would likely be **slightly right-skewed** (positively skewed) or nearly **symmetric**, as indicated by a skewness of 0.337.
- **Distribution**: With a mode of 12 and a mean of 14.10, we'd expect a more balanced distribution of scores with no extreme peak, as scores are spread out across a wider range.
- **Spread**: The higher standard deviation (3.573) and variance indicate more variability in this group, so the histogram bars would cover a broader range, extending from the minimum score of 9 to the maximum score of 20.
- **Kurtosis**: The negative kurtosis (-0.590) suggests a **platykurtic distribution**, meaning the histogram would be relatively flatter compared to a normal distribution, with a lower peak and thicker tails.

- **Interpretation**: The histogram would reflect a more diverse performance in the Non-mnemonic group, with scores spread across the spectrum, indicating less consistency in performance compared to the Mnemonic group.

Comparison of Histograms

- The **Mnemonic group histogram** would show a higher concentration of scores at the upper end (more scores around 18-20), with a sharp peak and a left-skewed shape.
- The **Non-mnemonic group histogram** would display a wider and flatter distribution, with scores more spread out from low to high and a slight right skew.

In summary, the Mnemonic group histogram would have a **sharp peak and left skew**, showing higher scores concentrated at the upper end, while the Non-mnemonic group histogram would be **flatter and more symmetric**, with scores dispersed over a wider range.

Activity 2

We conducted a study with 19 participants to test two different prototypes for the website https://www.snapfish.co.uk/photo-books where you can create and order a photo-book. Participants were asked to perform a number of tasks. One of the tasks required participants to upload 4 photos and then record the time (in seconds) in Prototype A (Pr.A), and then do the same for Prototype B (Pr.B). The data collected is provided below:

Participant/Case	Time of task in Pr.A	Time of task in Pr.B
1	30	65
2	60	**1400**
3	80	135
4	120	180
5	50	145
6	31	90
7	55	180
8	33	42
9	75	54
10	58	60
11	47	25
12	**800**	45
13	70	52
14	170	28
15	88	78
16	170	42
17	150	65
18	44	21
19	32	40

Attempt the following:

1. Form a research question that examines the difference in terms of time for Prototype A and Prototype B.

RQ: The UI design for uploading photos in snapfish to create a photobook has no effect on the time it takes to complete the task.

2. Form a hypothesis, which examines the difference between the two groups.

H_0: There is no significant time difference in the photo upload task to create a photobook in Snapfish between a group of users using Prototype A and Prototype B.

H_a: There is significant time difference in the photo upload task to create a photobook in Snapfish between a group of users using Prototype A and Prototype B.

3. What is the dependent variable?

The time to upload photos in Snapfish

4. What is the independent variable?

The use of prototype A or B

5. In SPSS or PSPP, set up a data file to record the above data. More specifically:
   - In **Variable View**, apply the appropriate Variable properties for the independent and dependent variables. (Tip: Look at how you did this the previous exercise).

   Look at Tutorial6.sav

   - In **Data view**, insert the above observations/data.
   - Have you got any outliers? What will you do about those?
   - Time of task in Pr.A – 800 and Time of task in Pr.B 1400 do not seem to be correct, the look like typing errors and you should correct those to 80 and 140 respectively that seams most likely to be a correct time.
   - Save the .sav file, as 'PhotoBookStudy'.

   Look at PhotoBookStudy.spv

6. Answer the following questions:
   a. Do we split the sample in this type of study according to treatment?

   No

   b. Is this type of study a 'between-subjects design' or a 'within-subjects design'? To answer the last question, perform a research using 'Google' regarding the terms: 'between-subjects design' and 'within-subjects design'.

   Within

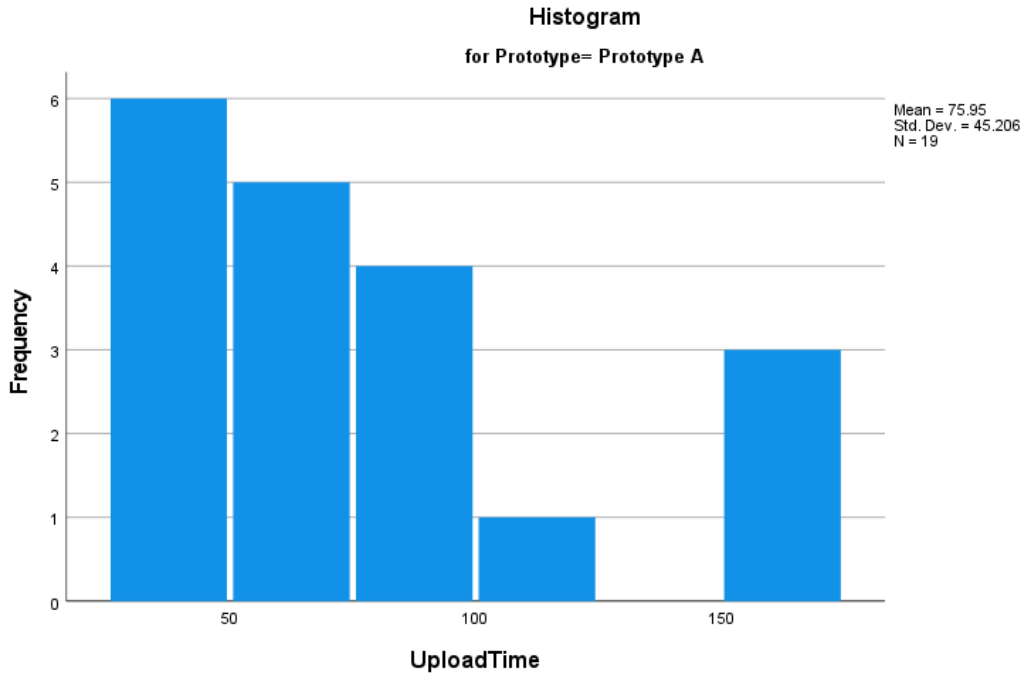
7. Use SPSS or PSPP to find the mean regarding the time for each prototype.
   a. Go Analyse>Descriptive Statistics>Explore
   b. In the Dependent List select the upload time, as factor list select the prototype used. In Statistics select Descriptives with Confidence Interval for Mean 95%. In plots select both Steam-and-leaf and Histogram. Select normality plots with tests. Look at the generated table and graphs, consult your lecture notes and discuss the output.
   c. Save it in the same folder as the PhotoBookStudy.sav (and with the same name).

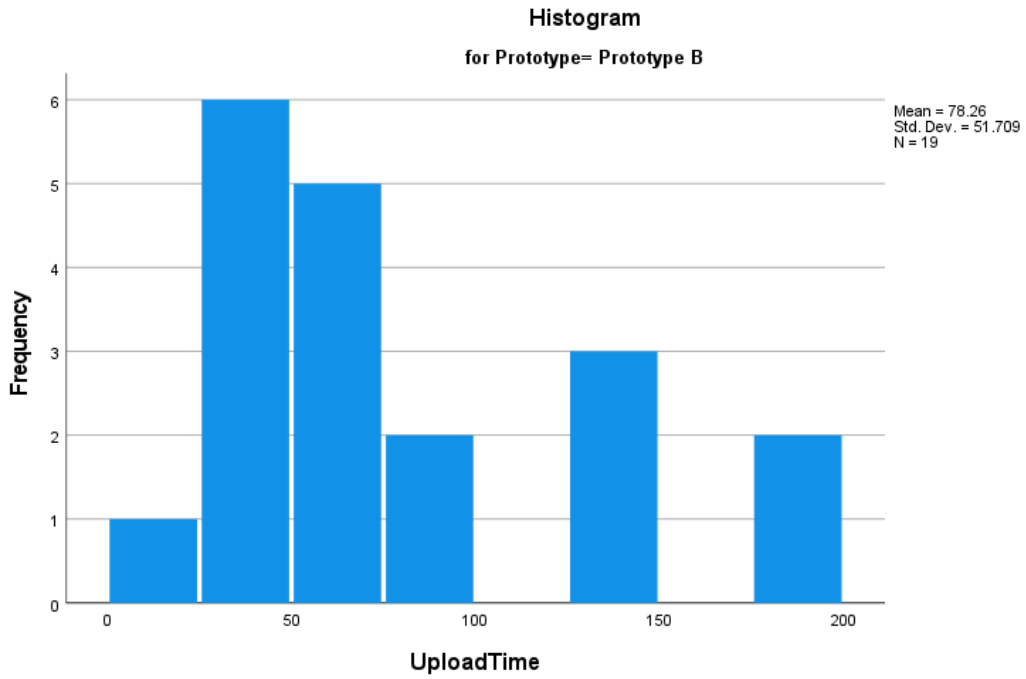
Descriptives

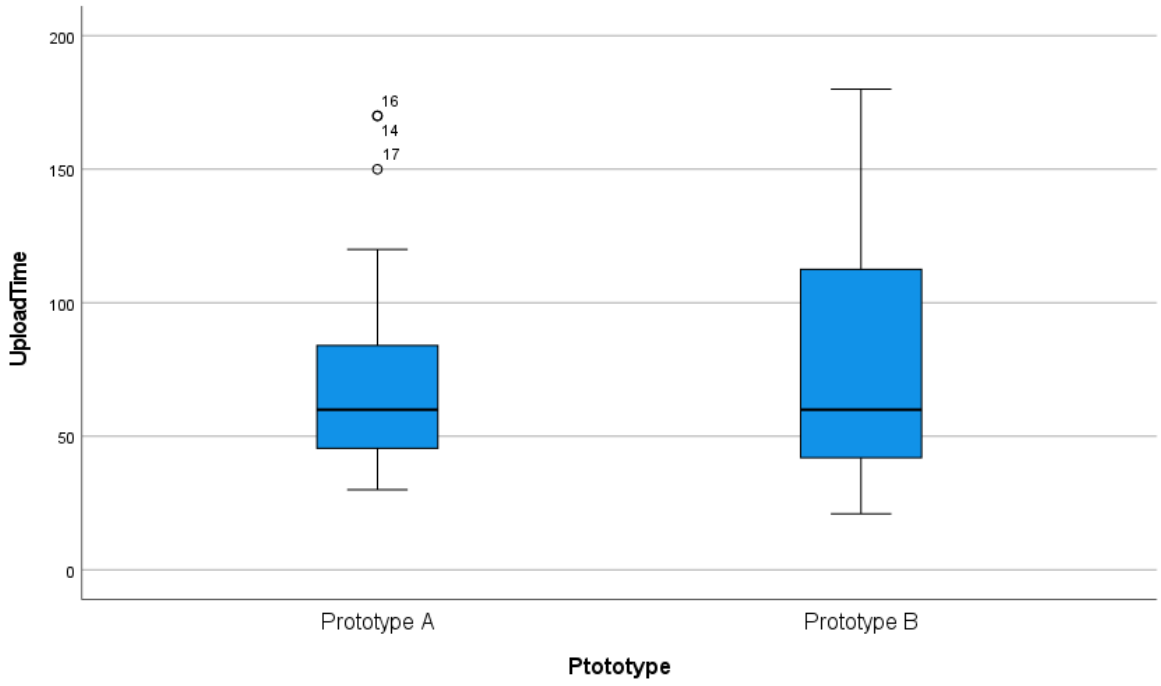
	Ptototype			Statistic	Std. Error
UploadTime	Prototype A	Mean		75.95	10.371
		95% Confidence Interval for Mean	Lower Bound	54.16	
			Upper Bound	97.74	
		5% Trimmed Mean		73.27	
		Median		60.00	
		Variance		2043.608	
		Std. Deviation		45.206	
		Minimum		30	
		Maximum		170	
		Range		140	
		Interquartile Range		44	
		Skewness		1.151	.524
		Kurtosis		.303	1.014
	Prototype B	Mean		78.26	11.863
		95% Confidence Interval for Mean	Lower Bound	53.34	
			Upper Bound	103.19	
		5% Trimmed Mean		75.79	
		Median		60.00	
		Variance		2673.871	
		Std. Deviation		51.709	
		Minimum		21	
		Maximum		180	
		Range		159	
		Interquartile Range		93	
		Skewness		.955	.524
		Kurtosis		-.404	1.014

Percentiles

			Percentiles						
		Ptototype	5	10	25	50	75	90	95
Weighted Average(Definition 1)	UploadTime	Prototype A	30.00	31.00	44.00	60.00	88.00	170.00	.
		Prototype B	21.00	25.00	42.00	60.00	135.00	180.00	.
Tukey's Hinges	UploadTime	Prototype A			45.50	60.00	84.00		
		Prototype B			42.00	60.00	112.50		


Histogram
for Prototype= Prototype A
Mean = 75.95
Std. Dev. = 45.206
N = 19
Frequency
0
1
2
3
4
5
6
50
100
150
UploadTime


Histogram
for Prototype= Prototype B
Mean = 78.26
Std. Dev. = 51.709
N = 19
Frequency
0
1
2
3
4
5
6
0
50
100
150
200
UploadTime




```
UploadTime Stem-and-Leaf Plot for
Prototype= Prototype A

 Frequency    Stem &  Leaf

     6.00        0 .  333344
     9.00        0 .  555677888
     1.00        1 .  2
     3.00 Extremes    (>=150)

 Stem width:  100
 Each leaf:        1 case(s)


UploadTime Stem-and-Leaf Plot for
Prototype= Prototype B

 Frequency    Stem &  Leaf

     7.00        0 .  2224444
     7.00        0 .  5566679
     3.00        1 .  344
     2.00        1 .  88

 Stem width:  100
 Each leaf:        1 case(s)
```

Think: After, finding the mean statistic, is there a significant difference between the two means?

There is a difference but we cannot say anything about the significance at this stage

Can we really judge how significant –how big- is their difference based on Descriptive statistics (i.e. calculation of the two means)?

No

Can we really answer the research question based only on the calculation of the means?

We can only make assumptions about the sample not for the population

Interpretation

1. **Central Tendency (Mean, Median, Trimmed Mean)**
    - The **mean upload time** is slightly higher for Prototype B (78.26) compared to Prototype A (75.95), suggesting Prototype B may take a bit longer on average.
    - The **medians** for both prototypes are the same (60.00), indicating that the midpoint of upload times is similar across both prototypes, despite the slight difference in means.
    - The **5% trimmed mean** (73.27 for A, 75.79 for B) suggests that outliers slightly impact the mean, particularly for Prototype B.
2. **Spread (Variance, Standard Deviation, Range, Interquartile Range)**
    - **Standard deviation** and **variance** are higher in Prototype B (SD = 51.709, variance = 2673.871) than in Prototype A (SD = 45.206, variance = 2043.608), indicating that Prototype B has more variability in upload times.
    - **Range** and **interquartile range (IQR)** are also higher for Prototype B (Range = 159, IQR = 93) compared to Prototype A (Range = 140, IQR = 44). This suggests that Prototype B has a wider spread of upload times, while times for Prototype A are more tightly clustered around the center.
3. **Skewness and Kurtosis**
    - **Skewness**: Prototype A has a skewness of 1.151, and Prototype B has a skewness of 0.955, both indicating a **positive skew** (right-skewed distribution). This means that for both prototypes, there are a few longer upload times pulling the distribution to the right.
    - **Kurtosis**: Prototype A has a kurtosis of 0.303 (slightly **leptokurtic**, meaning more peaked than a normal distribution), while Prototype B has a kurtosis of -0.404 (slightly **platykurtic**, meaning flatter than a normal distribution).
4. **Confidence Interval for the Mean**
    - The **95% confidence intervals** are fairly wide for both prototypes:
        - Prototype A: 54.16 to 97.74
        - Prototype B: 53.34 to 103.19

- These wide intervals reflect the high variability in upload times and suggest some uncertainty in the exact mean values.

Overall Insights

- **Prototype B** has a slightly higher mean upload time and greater variability, indicating it may be less consistent in performance compared to Prototype A.
- The **positive skew** in both prototypes suggests that most upload times are concentrated around the lower end, with a few longer upload times skewing the distribution.
- Given the wide **confidence intervals**, there may be overlap in performance between the prototypes, and further testing would be needed to confirm significant differences in mean upload times.

In summary, both prototypes exhibit similar median upload times, but Prototype B has a greater range and variability, which could indicate less predictable performance.

Interpretation of the histograms

Histogram for Prototype A

- **Shape**: The histogram for Prototype A would likely be **right-skewed** (positively skewed) due to the skewness value of 1.151. This means most upload times would be concentrated on the left side of the histogram, with a tail extending to the right.
- **Peak**: Given the median of 60 and a mean of 75.95, the histogram would show a peak around lower upload times, possibly between 50 and 70 seconds.
- **Spread**: The standard deviation (45.206) and range (140) suggest moderate variability in upload times, so the bars would extend over a broad interval but with most values clustered around the lower end.
- **Kurtosis**: With a kurtosis value of 0.303, the histogram would be **slightly leptokurtic**, meaning a relatively sharp peak around the mode and fewer extreme values compared to a normal distribution.
- **Overall Interpretation**: The histogram for Prototype A would show a concentration of upload times around the lower end, with a few longer times stretching to the right, indicating occasional higher upload times but a general tendency toward shorter times.

Histogram for Prototype B

- **Shape**: The histogram for Prototype B would also be **right-skewed** (positively skewed), with a skewness of 0.955, though less pronounced than Prototype A. This indicates a similar concentration of values on the left with a longer tail to the right, though the skew is less severe.
- **Peak**: With both a median of 60 and a mean of 78.26, the peak would likely be around 60 seconds, but the mean's slightly higher value indicates more spread toward longer times.
- **Spread**: Prototype B has a larger standard deviation (51.709) and range (159), so the histogram would be wider, indicating greater variability in upload times across participants.
- **Kurtosis**: The kurtosis of -0.404 suggests a **platykurtic distribution** (flatter than a normal distribution). This means the histogram would have a lower, broader peak and thicker tails, reflecting the spread of values.
- **Overall Interpretation**: The histogram for Prototype B would show a wider spread of upload times, with a peak around the lower end (around 60 seconds) but more frequent occurrences of longer upload times. This indicates less consistency in upload times for Prototype B compared to Prototype A.

Comparison of the Two Histograms

- **Prototype A**: The histogram would have a sharper peak with a more pronounced right skew, showing that most upload times are relatively consistent but with a few outliers on the higher end.
- **Prototype B**: The histogram would be flatter and wider, showing a more even distribution of upload times with greater variability, indicating that some users experienced significantly longer upload times.

In summary, the histogram for **Prototype A** would show more consistency and a sharp peak at lower upload times, while **Prototype B** would display a broader spread and less predictability in upload times.

Interpretation of Box Plots (or Steam and Leaf diagrams)

Box Plot for Prototype A

- **Median**: The median upload time for Prototype A is 60, so the line within the box would be at 60. This suggests that 50% of upload times are below 60 seconds and 50% are above it.
- **Interquartile Range (IQR)**: The IQR for Prototype A is 44, which indicates the middle 50% of data lies within a range of 44 units around the median. Given the median is 60, this suggests that the central data values are relatively close to each other, with the box spanning from around 38 to 82.
- **Whiskers (Minimum and Maximum)**: The minimum upload time is 30, and the maximum is 170, so the whiskers will extend from the bottom of the box down to 30 and from the top of the box

up to 170. However, due to the high maximum value compared to the IQR, the maximum may appear as an outlier, depending on the placement of the upper whisker.

- **Skewness**: Since Prototype A has a positive skew (skewness = 1.151), the box plot would likely show a longer upper whisker, indicating the presence of higher outliers or longer upload times.
- **Outliers**: There may be one or more individual points plotted above the upper whisker, representing outliers on the high end of the data (i.e., upload times much longer than typical for this prototype).

Interpretation for Prototype A: The box plot would show that most upload times are clustered around the lower end (close to the median of 60), with a few higher times causing a skew. The presence of a longer upper whisker and possible outliers suggests that, while most users experienced reasonably fast upload times, a few encountered much longer times.

Box Plot for Prototype B

- **Median**: The median for Prototype B is also 60, similar to Prototype A, so the line within the box would also be at 60, indicating the central tendency of the data.
- **Interquartile Range (IQR)**: The IQR for Prototype B is 93, which is more than twice as large as that of Prototype A. This wider box suggests that there is significantly more variability in the middle 50% of the upload times for Prototype B, indicating less consistency in performance.
- **Whiskers (Minimum and Maximum)**: The minimum is 21, and the maximum is 180, resulting in an even wider range (159) than Prototype A. The whiskers will extend from the lower end of the box to 21 and from the upper end of the box to 180. Similar to Prototype A, the upper whisker may be longer, possibly with outliers at the high end.
- **Skewness**: Prototype B is also positively skewed (skewness = 0.955), though less so than Prototype A. This would likely be reflected by a slightly longer upper whisker, but not as extreme as in Prototype A.
- **Outliers**: Prototype B may also have outliers, as indicated by the high maximum value (180), which is far above the median. These outliers reflect the inconsistency in upload times, with some users experiencing significantly higher times.

Interpretation for Prototype B: The box plot for Prototype B would show a broader spread of upload times, with a wider box and potentially longer whiskers. This indicates greater variability and less consistency in upload times, with some users experiencing very high values. The wider IQR suggests a larger spread in the typical experience for Prototype B users compared to Prototype A.

Comparison of the Box Plots for Prototype A and Prototype B

- **Central Tendency**: Both prototypes have the same median (60), suggesting that the midpoint of upload times is similar. However, the spread and variability around this median are different.
- **Variability**: Prototype B shows a much wider IQR and range, indicating more inconsistency and a broader distribution of upload times. Prototype A has a smaller IQR, meaning the central 50% of its data is more tightly clustered around the median.
- **Skewness and Outliers**: Both prototypes are positively skewed, but Prototype A shows a higher skewness, likely with a more pronounced upper whisker or more outliers at the higher end.

In summary, **Prototype A** has a more consistent upload time with less variability, while **Prototype B** shows greater variability and more extreme values. This suggests that Prototype A may provide a more predictable user experience in terms of upload times, whereas Prototype B has a wider range of user experiences, with some users experiencing significantly longer upload times.

Activity 3

Read and reflect on an interesting article about mean and mode

Mean is the most popular way to calculate the average, even when it is not appropriate to use it (e.g. when calculating the average of ordinal variables such as 'number of children a woman has in UK'). However, read an article where mode gives us more interesting and valuable information: 'Small Data: When average age of death was zero' (source: http://www.bbc.co.uk/news/blogs-magazine-monitor-29618190)